



JOHN THE REAPER



Ultimate Jumbo Hash Harvesting Cheat-Sheet • Under the Moonlight

SECTION I: CORE SYNTAX & EXECUTION PRINCIPLES

Standardize invocation parameters to maximize thread throughput and bypass unnecessary detection cycles.

```
# Basic Invocation Structure
john [options] [path_to_hash_file]

# Automatic Heuristic Detection & Cracking
john --wordlist=/usr/share/wordlists/rockyou.txt hashes.txt

# Forced Format Execution (Prevents misidentification & speeds up cracking)
john --format=[format_code] --wordlist=[path_to_wordlist] [path_to_file]

# Real-World Example: Raw MD5 Hash
john --format=raw-md5 --wordlist=/usr/share/wordlists/rockyou.txt hash.txt
```

 **REAPER'S MARK:** Standard cryptographic hashes (e.g., plain MD5, SHA1, SHA256) strictly require the **raw-** prefix in John Jumbo builds (e.g., **raw-md5**, **raw-sha256**, **raw-sha512**).

SECTION II: FORMAT LISTING & HASH IDENTIFICATION

When an unknown hash remains silent under autodetect, query internal signatures or invoke offline probes.

```
# List supported raw/dynamic formats
john --list=formats | grep -iF "md5"

# Display detailed subformats and configurations
john --list=subformats

# External Probes: Offline hash identification tools
python3 hash-id.py
nth -t "5b31f93c09ad1d065c0491b764d04933"
```

SECTION III: SESSION CONTROL & OUTPUT MANAGEMENT

```
# View harvested cleartext passwords stored in ~/.john/john.pot
john --show --format=[format_code] hash.txt

# Resume a paused or interrupted cracking session
john --restore
```

Interactive Live Session Controls (While Running)

- **[Spacebar]** / **[Enter]**: Displays current execution metrics, candidate rate (p/s), and ETA.
- **Ctrl + C**: Suspends execution safely and flushes session progress to disk.
- **q**: Aborts the cracking process immediately.

🕯️ SECTION IV: REAPING OS AUTHENTICATION HASHES

1. Windows Hashes (NTLM / SAM Database)

```
# Extracted via Mimikatz or secretdump.py
john --format=NT --wordlist=/usr/share/wordlists/rockyou.txt ntlm_hashes.txt
```

2. Linux Hashes (/etc/passwd + /etc/shadow)

```
# Step 1: Bind identity and shadow files together
unshadow /etc/passwd /etc/shadow > unshadowed.txt

# Step 2: Slice through Linux SHA-512 crypt hashes
john --format=sha512crypt --wordlist=/usr/share/wordlists/rockyou.txt unshadowed.txt
```

🧑🏻 SECTION V: SHADOW MANGLING & SINGLE MODE

📖 Single Crack Mode (--single)

Exploits weak passwords derived from usernames, GECOS fields, or system identity context.

File Format Requirement: Target hashes must prepend the corresponding username.

```
mike:1efee03cdcb96d90ad48ccc7b8666033
```

```
# Execute Single-Mode attack
john --single --format=raw-md5 hashes.txt
```

👤 Custom Mangling Rules (john.conf)

Define custom rules in `/etc/john/john.conf` or `/opt/john/john.conf`:

```
[List.Rules:PoloPassword]
cAz"[0-9] [!£$%@]"
```

- **c** : Capitalize first character (`polo` → `Polo`).
- **Az** : Append defined characters to the end.
- **A0** : Prepend defined characters to the beginning.
- **[0-9]** : Iterate digits 0 through 9.
- **[!£\$%@]** : Iterate specified special symbols.

```
# Execute with custom rule set
john --wordlist=/usr/share/wordlists/rockyou.txt --rule=PoloPassword hash.txt

# Dry-Run Rule Preprocessor (Test rules without cracking)
john --wordlist=test.txt --rule=PoloPassword --stdout
```

🔑 SECTION VI: THE CONVERSION SUITE (*2JOHN)

Convert binary containers and private key formats into standard John Jumbo hash strings.

TARGET TYPE	CONVERSION COMMAND	HARVESTING COMMAND
Linux Shadow	<code>unshadow /etc/passwd /etc/shadow > hash.txt</code>	<code>john --wordlist=rockyou.txt hash.txt</code>
ZIP Archive	<code>zip2john file.zip > zip_hash.txt</code>	<code>john --wordlist=rockyou.txt zip_hash.txt</code>
RAR Archive	<code>rar2john file.rar > rar_hash.txt</code>	<code>john --wordlist=rockyou.txt rar_hash.txt</code>
SSH Private Key	<code>python3 /usr/share/john/ssh2john.py id_rsa > ssh_hash.txt</code>	<code>john --wordlist=rockyou.txt ssh_hash.txt</code>
PDF Document	<code>pdf2john.pl document.pdf > pdf_hash.txt</code>	<code>john --wordlist=rockyou.txt pdf_hash.txt</code>
KeePass DB	<code>keepass2john database.kdbx > keepass_hash.txt</code>	<code>john --wordlist=rockyou.txt keepass_hash.txt</code>

⚡ SECTION VII: NIGHT SPELLS & OPTIMIZATIONS

🌙 The `john.pot` Urn

- **Location:** `~/.john/john.pot`
- **Function:** Stores all successfully cracked hashes in `hash:plaintext` format.
- **Troubleshooting:** If John exits with *0 password hashes cracked left*, the hash was already cracked. View it using `john --show hash.txt`.

🚀 Multi-Core Parallel Processing

```
# Fork processes across multiple CPU threads
john --fork=4 --wordlist=/usr/share/wordlists/rockyou.txt hash.txt
```

🕯️ Pure Brute-Force (Incremental Mode)

```
# Exhaust character permutations when no wordlist is supplied
john --incremental --format=raw-md5 hash.txt
```